

The AgentOps Operating Model

Evals, tracing, and regression tests for AI agents — one loop, before, at, and after ship.

OBSERVE · EVALUATE · GATE · CONTROL

If it can act, it needs traces, tests, gates, and a kill switch.

	BEFORE SHIP	AT SHIP	AFTER SHIP
Observe the map	Instrument OpenTelemetry GenAI semantic conventions Structured spans: step, tool, params, outputs, errors	Capture the release trace Every span tagged with agent version + instructions hash Breadcrumbs, not brain soup	Traces in Application Insights + Foundry GA End-to-end: model calls and tool actions Replay the decision path, not the answer
Evaluate the anchors	Golden set Four categories: happy path, edge, pushback, adversarial Threshold per category, not one global score	Offline eval per version GA Rule-based scorers for tool use; LLM judge for judgment Built-in agent evaluators + custom code evaluators	Continuous evaluation on sampled traffic GA Scores land beside the traces Drift = trend against a pinned baseline
Gate the rope	PR trigger Any change to prompts, knowledge, tools, or evals runs the matrix Fail the build, not the user	Matrix + environment-approved promotion Candidate vs. current side by side A human approves the route change; eval and deploy identities are separate	Azure Monitor thresholds Cost, latency, eval scores, safety flags Five to seven signals. Watch them daily.
Control the way out	Define before you need it Immutable versions; the last good one stays warm Rollback trigger written down, with an owner	Route change, not redeploy Promotion and rollback are the same operation Shadow and canary before broad production	Three levels of stop Tool, agent, fleet Playbook: triage → contain → roll back → learn

Seven things to version

They touch every cell.

- 01 **Model version**
- 02 **Prompt / instructions**
the demo's regression
- 03 **Tool contracts**
schema + MCP server version
- 04 **Evaluator version**
the ruler — version it too
- 05 **Policies / guardrails**
- 06 **Retrieval index**
- 07 **Agent configuration**
temperature, limits, routing

An agent release = model + instructions + tool schema + evaluator version.
Pin all of it. Tag every trace with it.

GA generally available in Microsoft Foundry
PREVIEW public preview
Untagged cells are vendor-neutral practice.

Monday morning — the starter kit, not the cathedral

- ☐ Add tracing with structured spans
- ☐ Capture tool calls: inputs, outputs, errors
- ☐ Write down twenty cases in four categories
- ☐ Add ONE regression check to CI
- ☐ Choose one irreversible action that needs a human
- ☐ Define the rollback trigger and who can pull it
- ☐ Watch token cost daily